Introduction to Computer Hardware

Contents

- Computer Hardware Components

  Casing, Processor, Motherboard,

  RAM, Power supply
- How to assemble a PC

Hardware Components

Hardware & Software

Hardware

All of the electronic and mechanical equipment in a computer is called the hardware. Examples include:

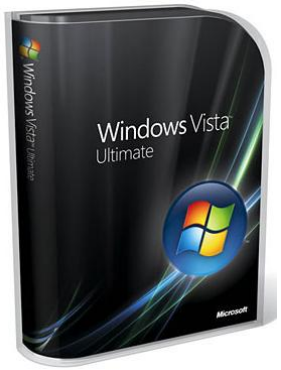
- **Motherboard**
- **Hard disk**
- **RAM**
- **Power supply**
- **Processor**
- **Case**
- **Monitor**
- **Keyboard**
- **Mouse**

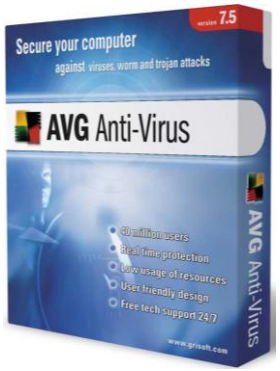
Hardware & Software

Software

The term software is used to describe computer programs that perform a task or tasks on a computer system. Software can be grouped as follows:

- System software - Operating System etc.
- Utility programs - BIOS, Partition, Antivirus etc.
- Applications Software - Word, Photoshop, etc.





PC Components

Computer system - collection of electronic and mechanical devices operating as a unit. The main parts are:

1. System unit
2. Monitor
3. Keyboard
4. Mouse
5. Speakers

System Unit

The **system unit** is the main container for system devices. It protects the delicate electronic and mechanical devices from damage. Typical system unit devices include:

- Motherboard
- CPU (Processor)
- Memory
- Disk drives
- Ports - USB etc.
- Power supply
- Expansion cards - sound card, network card, graphics card etc.

Peripherals

Peripherals are devices that connect to the system unit using cables or wireless technologies. Typical peripherals include:

- Monitor
- Keyboard
- Printer
- Plotter
- Scanner
- Speakers

Plotter

The Computer System

Input Devices

- Key board
- Mouse
- Microphone
- Scanner
- Web Cam

- MODEM
- Touch Screen Monitors
- USB Pen drives
- CD/DVD RW
- Floppy Drive

Mother Board
Processor
RAM
VGA Adaptor
Hard Disk
Power Supply

System Unit

Output Devices

- Monitor
- Printer
- Speakers

- MODEM
- Touch Screen Monitors
- USB Pen drives
- CD/DVD RW
- Floppy Drive

Basic Component Of a Computer System Unit

- Casing
- Power Supply
- Mother Board
- Processor
- Memory (RAM)
- Display Adaptor
  - AGP
  - PCI Express
- Storage devices
  - Floppy disk
  - Hard Disk
  - CD/DVD ROM
- Sound Card
- Network Interface
- Monitor /Keyboard/Mouse
- Operating System

Casing

- Outer cover
- Two types:
  - Desktop type
  - Tower type
- Ventilation is very important.

Desktop type casing

Tower type casing

Types of cases

Full Tower

Server Blade

Mini Case

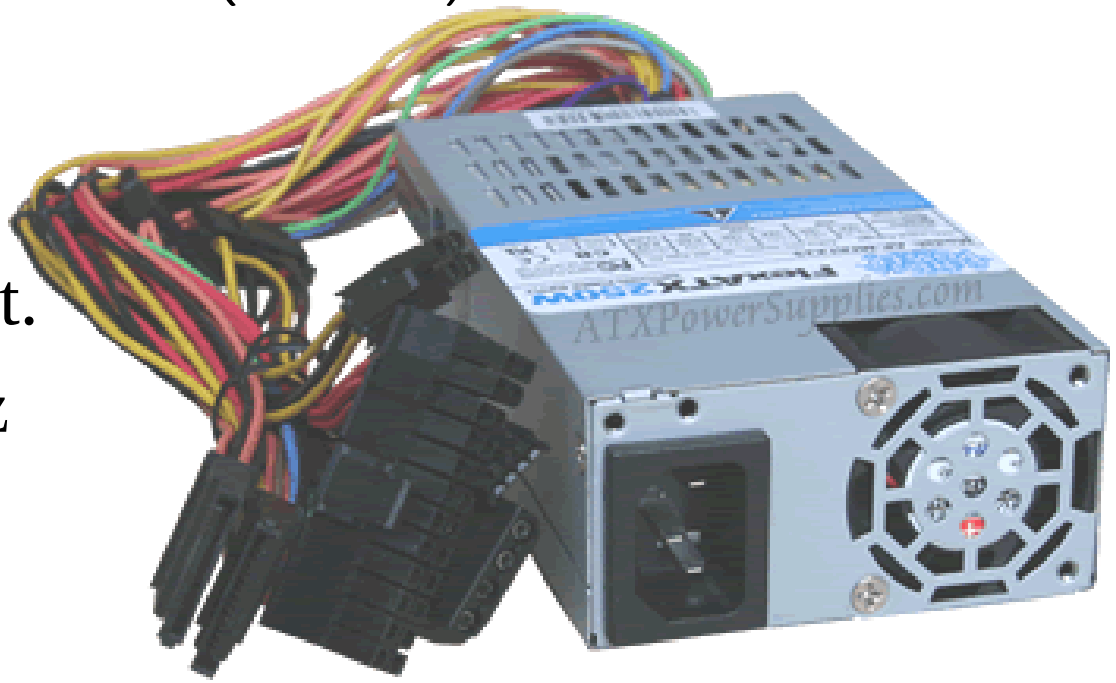
Mid Tower

Micro case

Selecting a casing

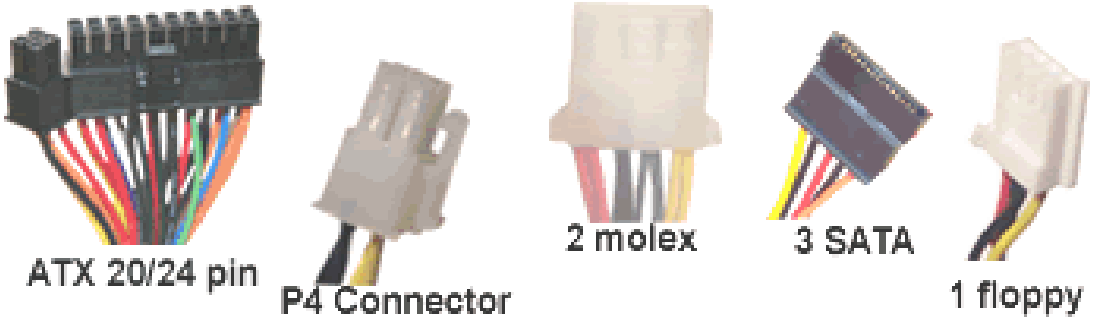
- Airflow (Limited airflow leads to overheating)
- Cooling Systems (with extra fan)
- Front-panel Connectivity (USB..)
- Make sure that the cabinet has enough space (adding additional h/w)

Power Supply Unit (PSU)



- Power distribution unit.
- Inputs AC-240V/60Hz
- The connector types available
- Wattage is very important
  - 350W
  - 450W

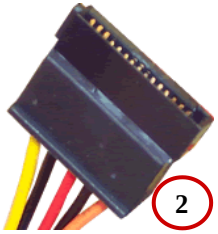
Connectors included:

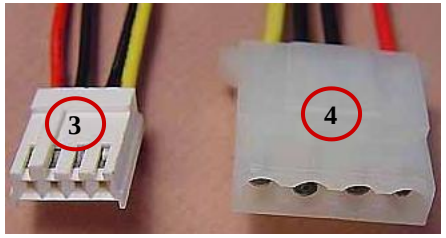


Power Supply Unit (PSU)

- Uses advances power management (**APM**) to allow the computer go into a standby mode
- Some have a switch to toggle between 240 volt supplies and 110 volt supplies.

- The main connections are:

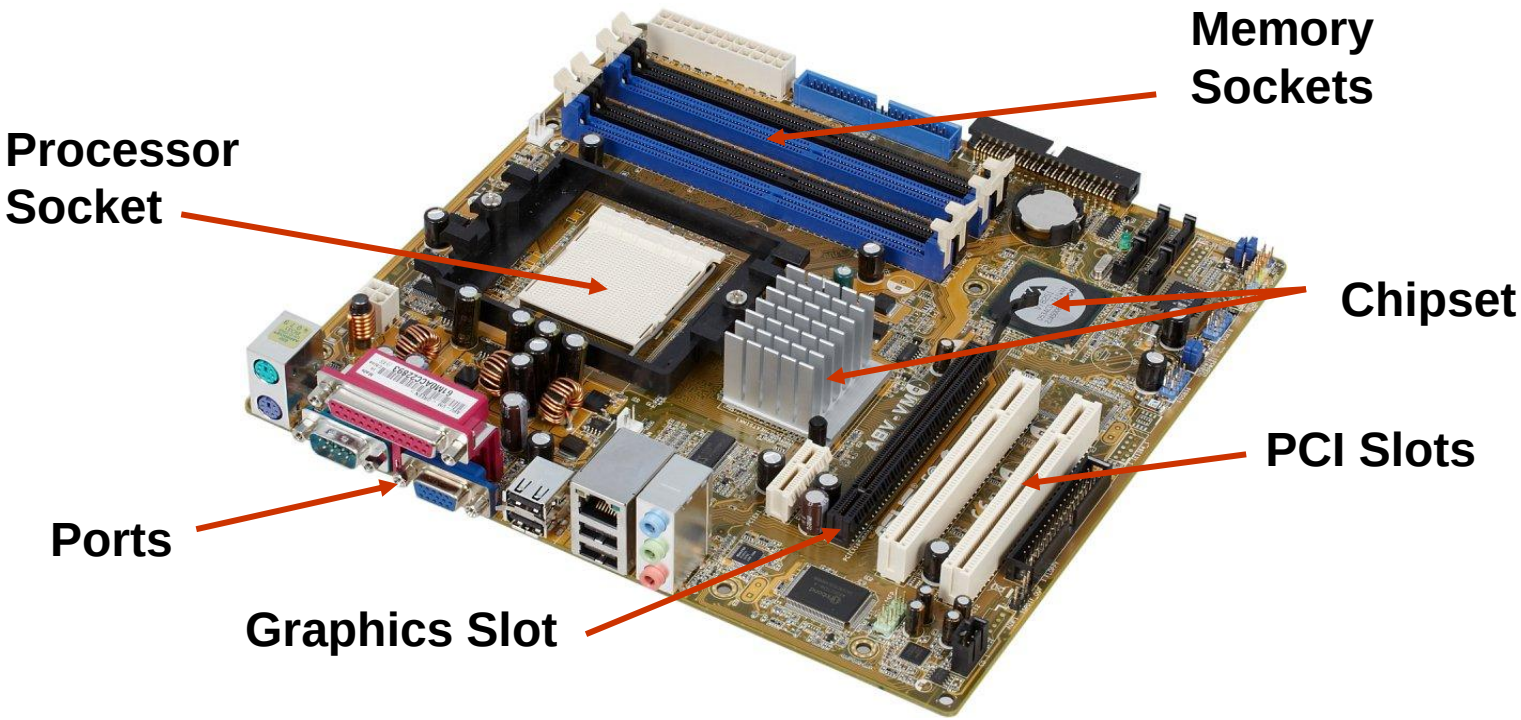




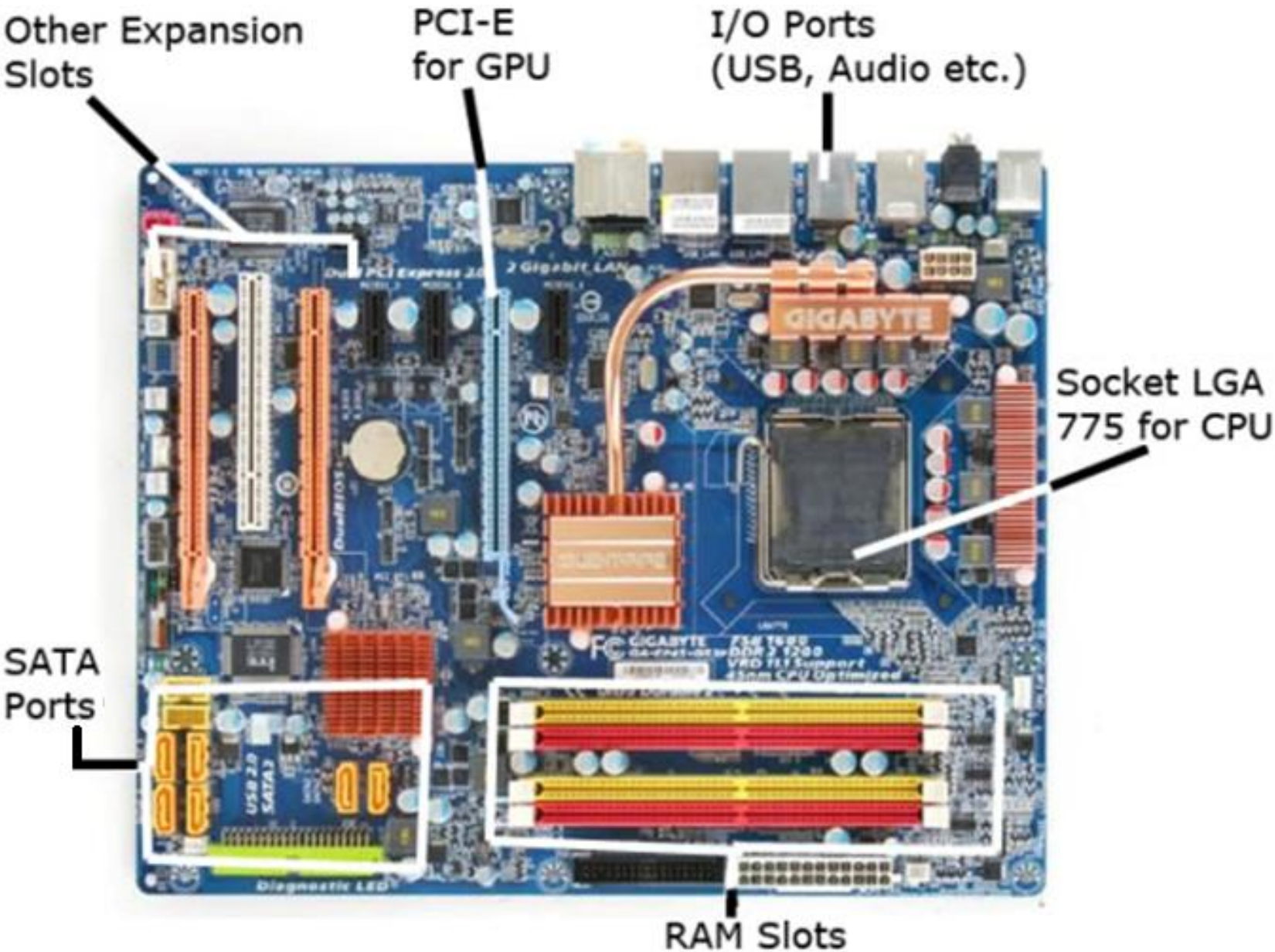
1	**Main connector**	**Connects to the motherboard and supplies the 3.3 and 5 volt supply for the board.**
2	**SATA connector**	**Connects SATA drives**
3	**Berg connector**	**Connects floppy disk drives**
4	**Molex connector**	**Connects IDE hard drives and optical drives.**

Motherboard

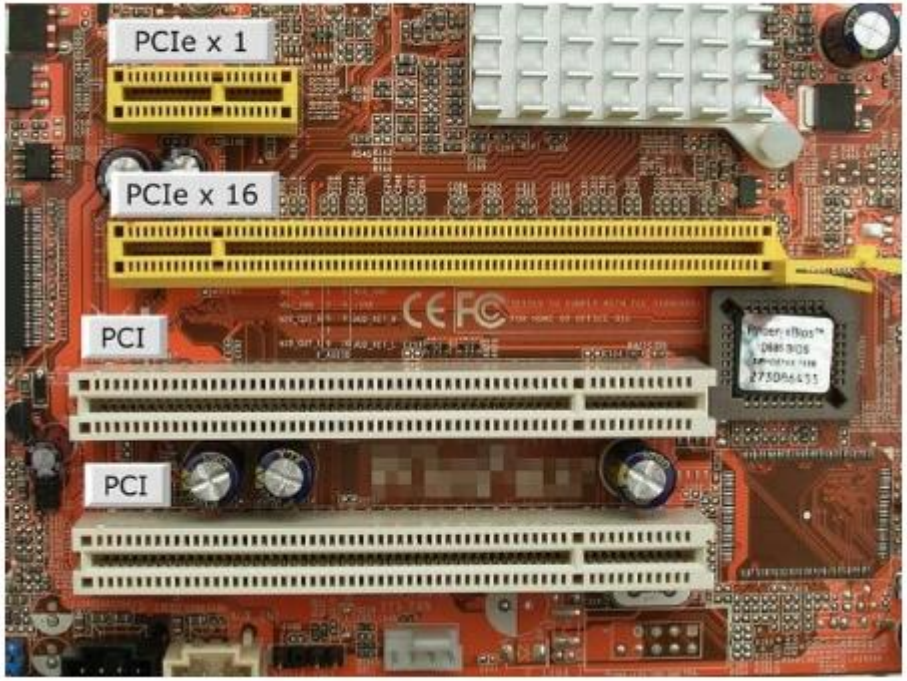
Mainboard or **system board** - the main circuit board for the computer system. All device in the computer system will either be part of the motherboard or connected to it.



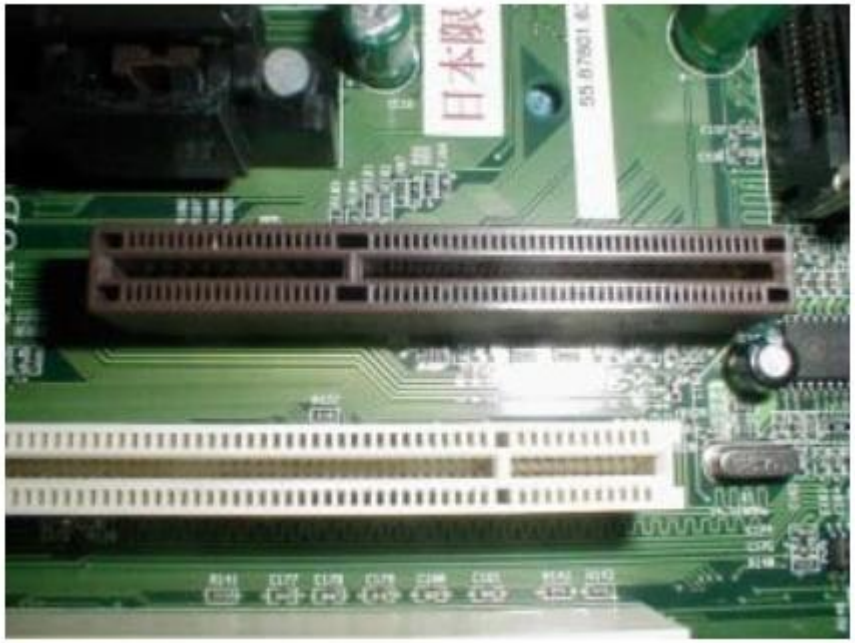
Motherboard



PCI-e slot (x16)



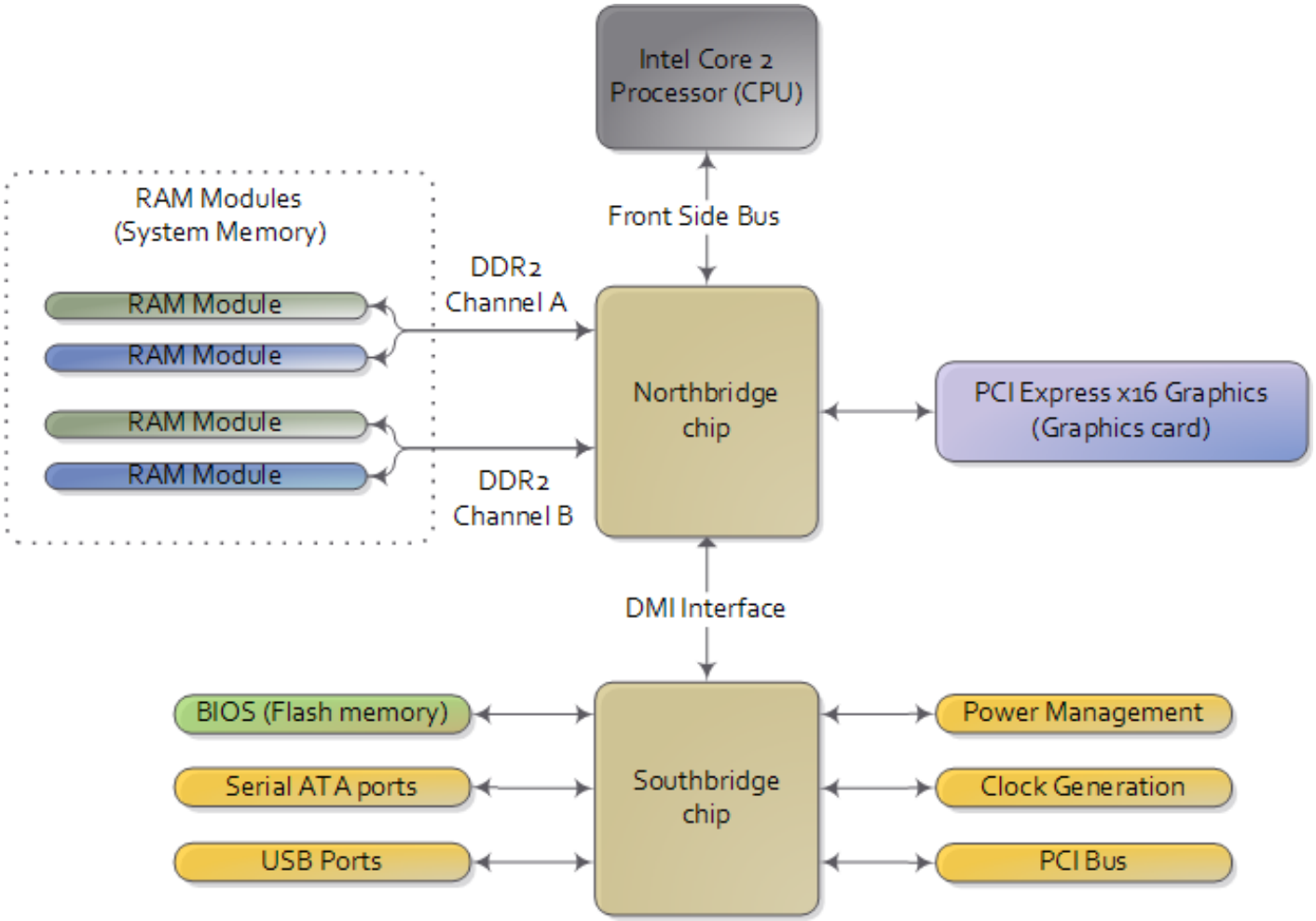
AGP slot



PCI Express: The best type of expansion slot to have in your PC is the PCI Express, also written as PCIe

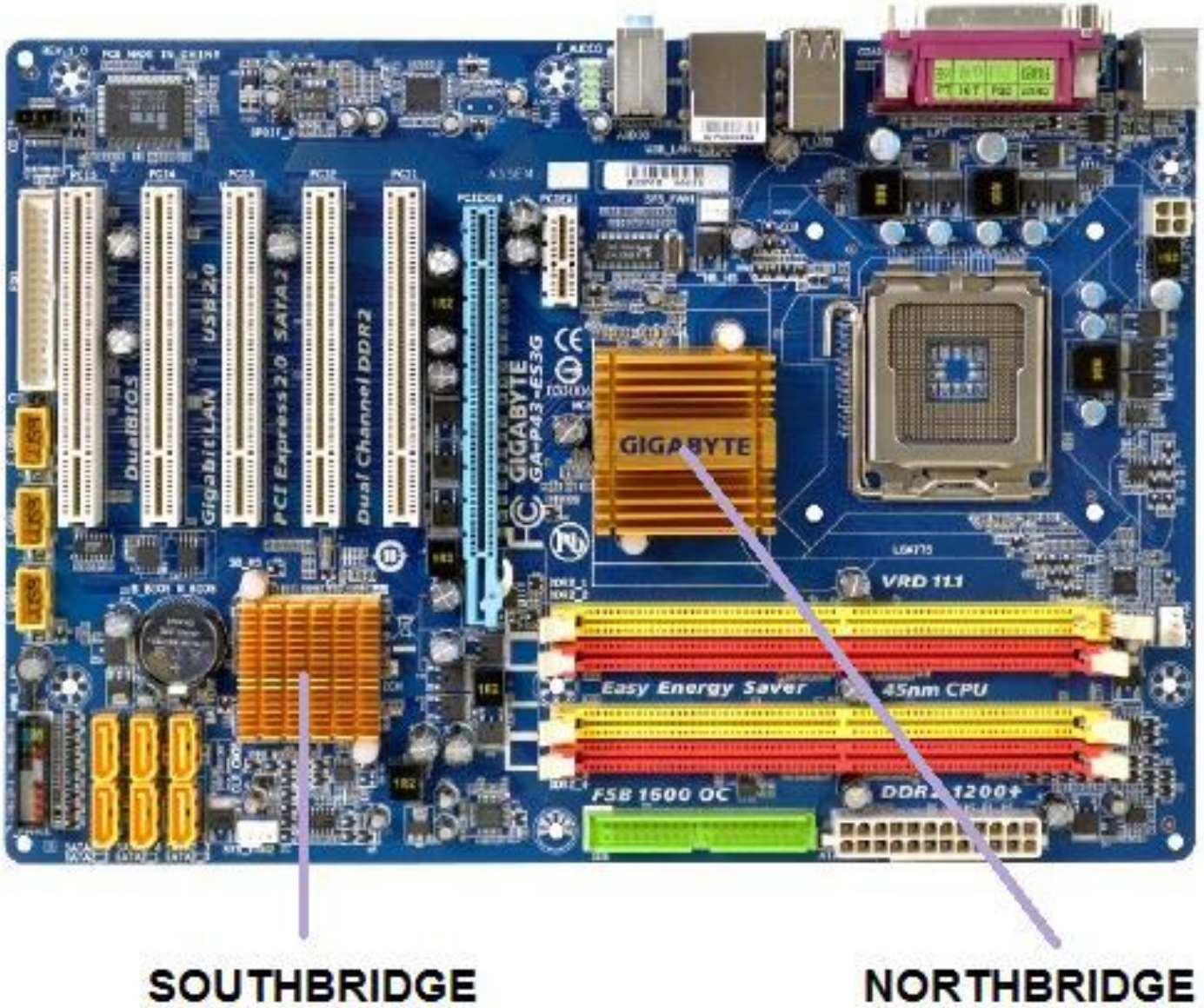
AGP: Was specifically designed to deal with graphics adapters.

Chipset of MB



The chipset is a silicon backbone integrated into the motherboard that works with specific CPU generations. It relays communications between the CPU and the many connected storage and expansion devices.

Chipset of MB



ROM-BIOS/POST

- **Basic Input / Output System (BIOS)**
  - boot the computer by providing a basic set of instructions
  - perform POST (Power-On Self Test)
  - load the operating system after turning on the computer
  - provide set-up program for changing BIOS parameters
  - provide low-level routines for the communications between OS and hardware devices

BIOS is stored on the ROM chip

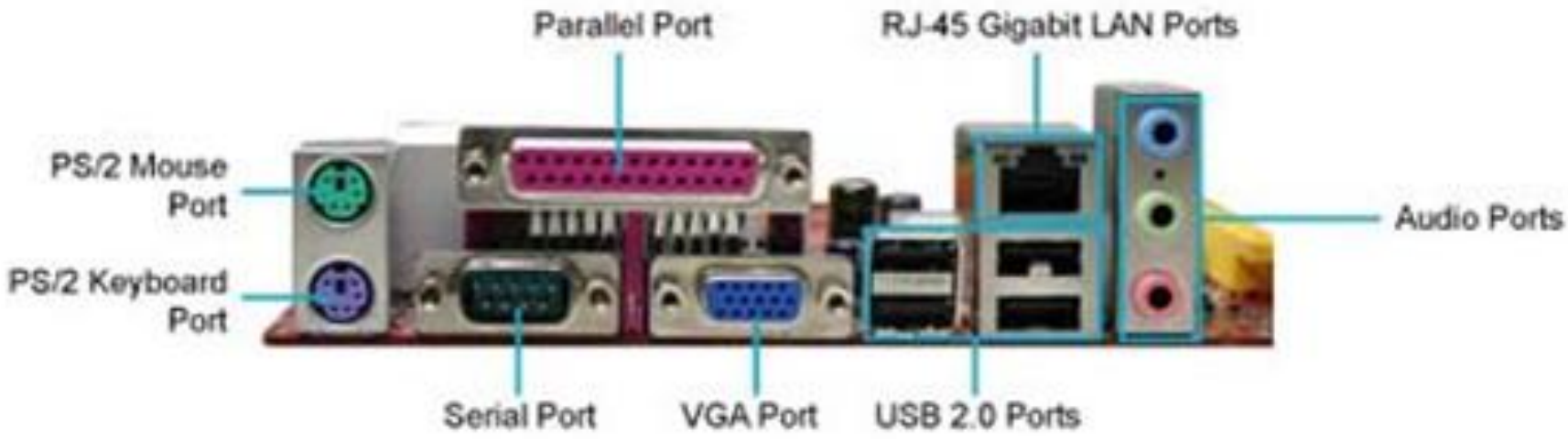
Ports

Computer ports are interfaces between peripheral devices and the computer. They are mainly found at the back of the computer but are often also built into the front of the computer chassis for easy access.

Ports at the rear of the computer

Ports at the front of the computer

Back panel of a Motherboard



- **Serial port** - a 9-pin port. Often called Com ports - Com1, Com2 etc. Mice and external modems were connected to these ports.

- **Parallel port** - a 25-pin port used to connect printers, scanners, external hard disks, zip drives etc. to the computer. They are often called LPT ports - LPT1, LPT2 etc.

LPT (line print terminal)

- **Video port -** used to connect a monitor to the computer system. There are two types:
  - **VGA port** - This is a 15-pin port and is **blue** in colour. It is an analogue port and is being replaced by the DVI port.

  - **DVI port** - **white** in colour, it is a digital port. This means that no conversion is necessary between the computer and the monitor and that means that images can be produced more quickly on the monitor

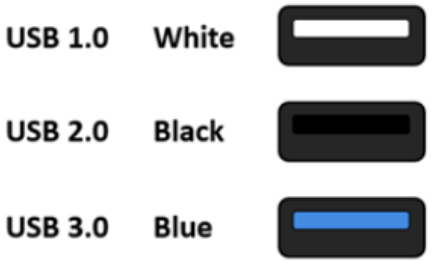
- **PS/2 port** - used to connect keyboards and mice to the computer. The keyboard port is **purple** and the mouse port is **green**

- **Modem port** - used to connect a modem to a telephone line. **RJ11** is the technical term for the port.

- **USB port** - intended to replace Serial, Parallel and PS/2 ports with a single standard. 127 devices can be connected to a single USB port. **Hot swappable** - devices can be connected and disconnected while the computer is on

There are different USB standards in use:

- **USB 1** - original standard - transfer data of **1.5MBps**.
- **USB 2** - current standard - transfer rate of **60MBps**.
- **USB3** - future standard (2009) transfer rate of **600MBps**.



- **FireWire port -** (IEEE 1394) is an Apple technology There are two versions available and a third is planned:

- **FW 400** - transfer rate of **50MBps**.
- **FW 800** - transfer rate of **100MBps**.
- **FW S3200** - transfer rate of **400MBps**.

- USB devices must be connected to a host computer while FireWire devices can be connected to each other without using a computer.

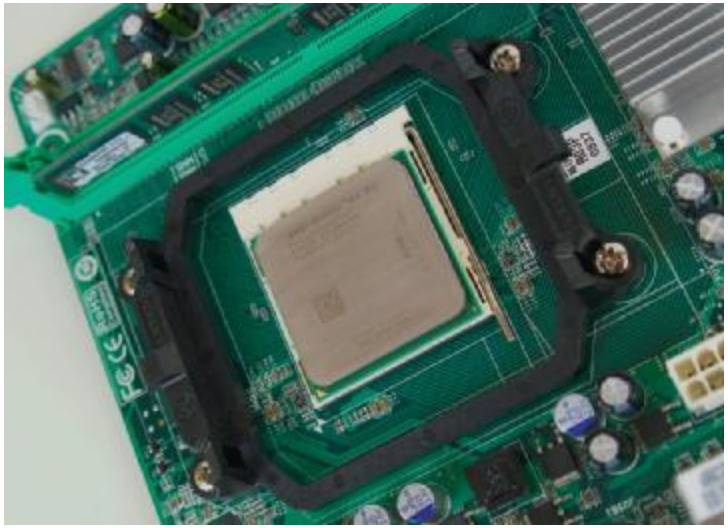
- **Ethernet port** - used to connect to a network. Known as **RJ45**, it is larger than a modem port.

- **Audio ports** - used to input and output audio from the computer. Three mini jack ports but there may be more:

  - **Light blue** - Line in - connect external devices
  - **Lime** - Connect the speakers to this port.
  - **Pink** - Connect a microphone to this port.

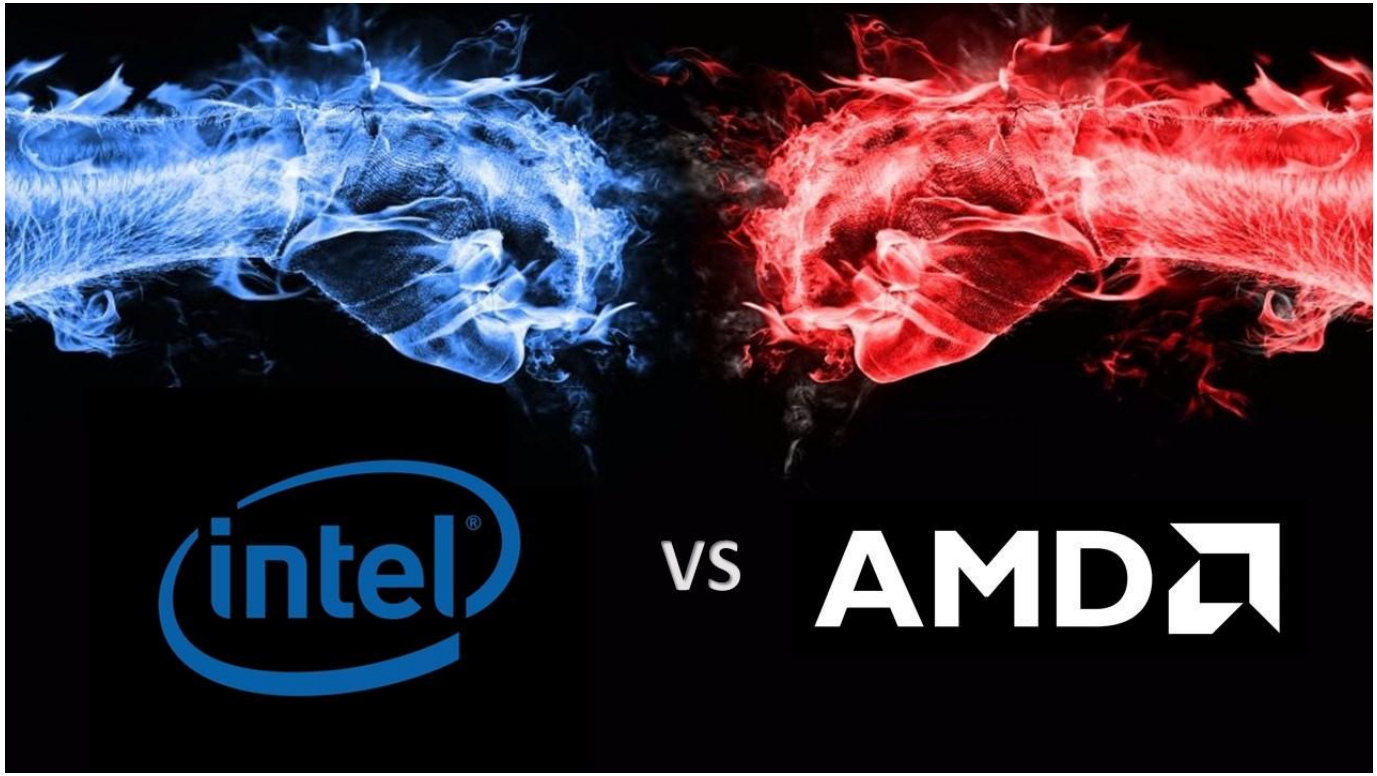
PROCESSOR

- The processor installed on a motherboard is the primary component that determines the computing power of the system
- Two major manufacturers of processors are:
  - Intel (*www.intel.com)*
  - *AMD (www.amd.com)*



PROCESSOR

Intel : AMD



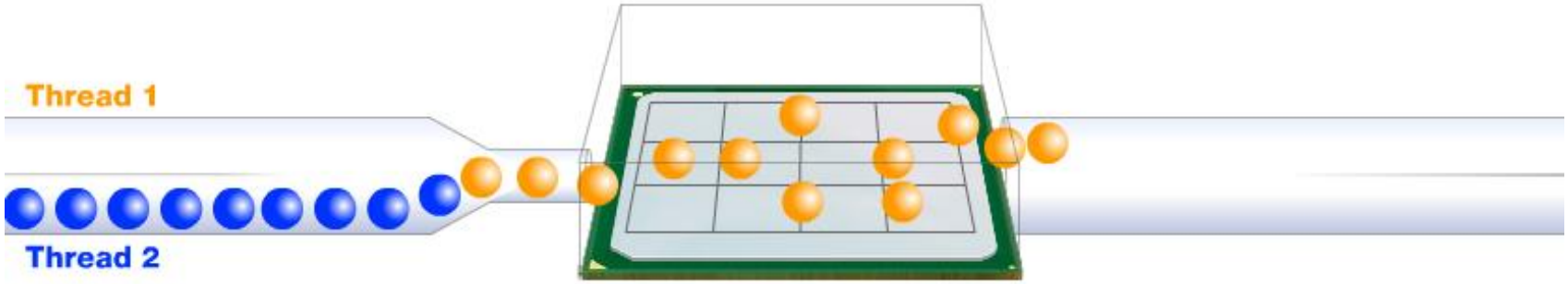
PROCESSOR

Discuss the evolution

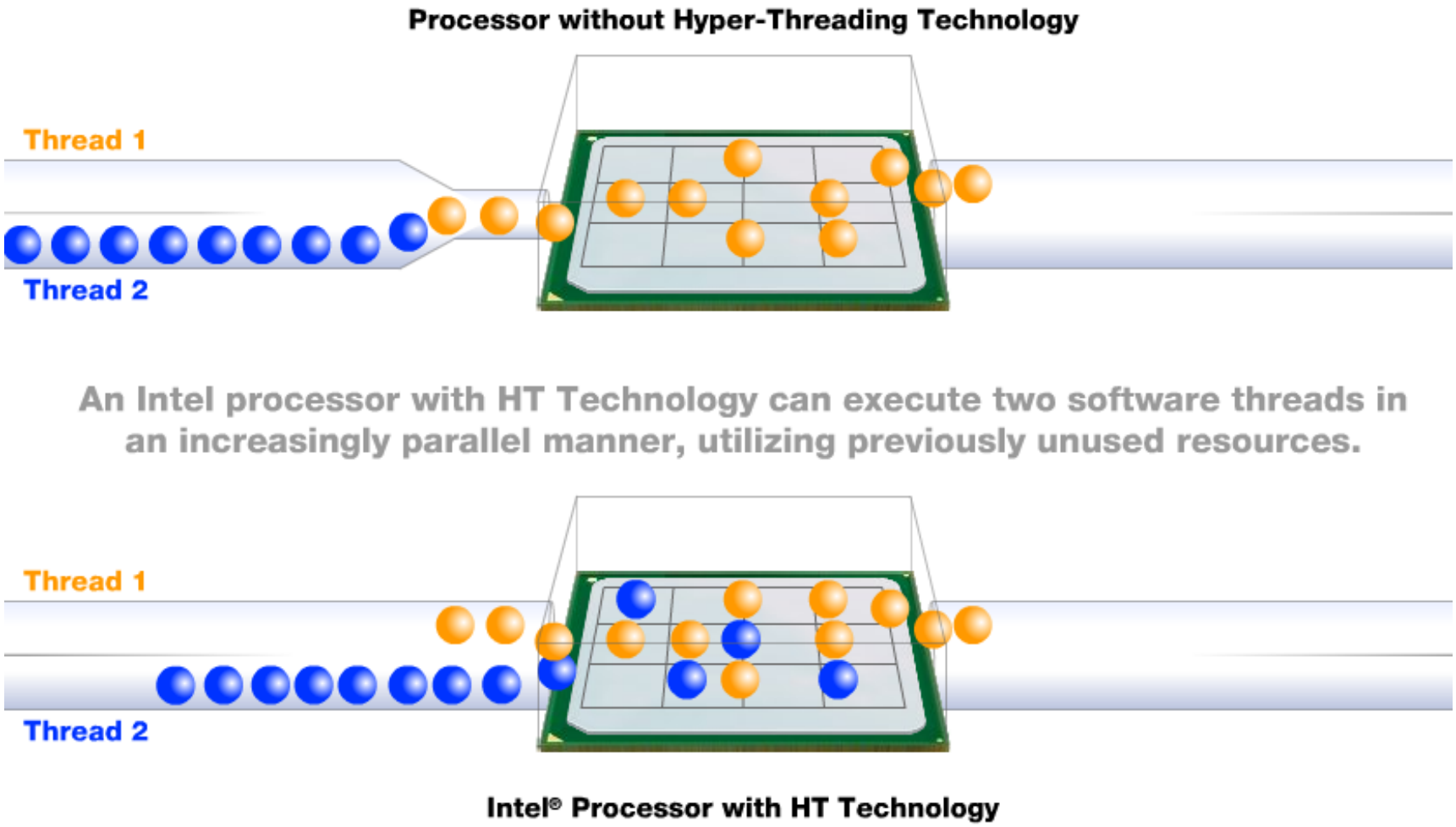
Intel : AMD

PROCESSOR

Single-Core
Processors
At times, while instructions are being carried out, additional information has to "wait in line" for the processor's resources.


Thread 1
Thread 2

PROCESSOR


Processor without Hyper-Threading Technology
Thread 1
Thread 2
An Intel processor with HT Technology can execute two software threads in an increasingly parallel manner, utilizing previously unused resources.
Thread 1
Thread 2
Intel® Processor with HT Technology

PROCESSOR

Dual-Core
Processors

The new Intel® Pentium® D processor features a "dual-core" processor, which means there are ***two full*** processor ***cores*** built into a single package.

Question ?

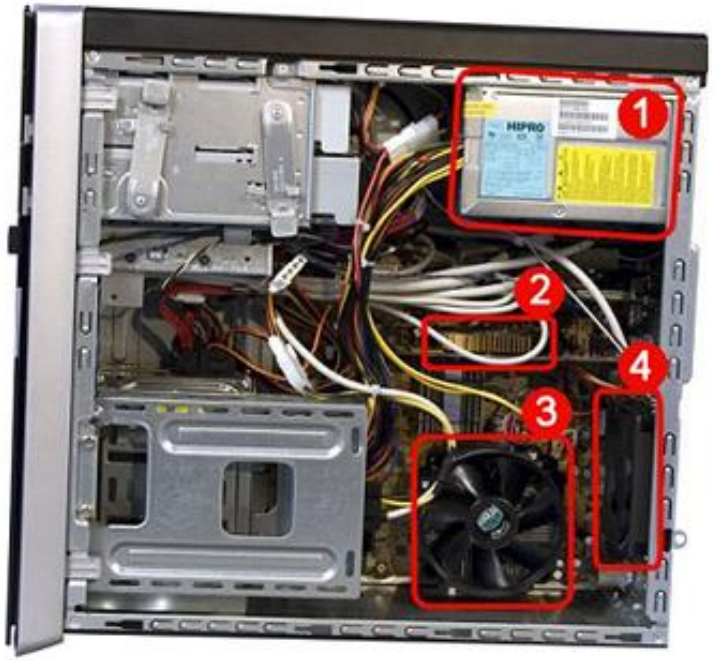
Over the years, the clock rate of computers has increased, rising from 1 MHz in the 1970s to 4 GHz today. However, over the last few years, clock speeds have not increased greatly. Briefly explain why the clock speed of modern processors is limited, and state what engineers are doing to overcome this limitation.

A processor’s energy consumption increases with clock speed. Consequently, the processor gets hotter at higher speeds. Since the size of the chip is only a few millimetres square, the heat can only be dissipated by large heat sinks. Now the limit as exceeded by using heat sinks and cooling fans.

Another limitation on processor speed is memory access times. The rate at which processors have increased speed has not been matched by a corresponding increase in memory access times. Therefore, memory access is proving to be a bottleneck and limits the performance of a computer system.

Multicore: The multicore processor has 2,3,4 or more CPUs on the same chip. Each CPU can process data in parallel. This allows more computations per second by executing them in parallel.

Water cooling - Liquid cooling is one way to cool a CPU because water transfers heat much more efficiently than air. Liquid cooling also makes your PC run quieter because you won’t have fans constantly running at a high RPM. However, liquid cooling a PC can also be dangerous if water leaks onto hardware.


1
2
3
4

5
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fractal
design
650W

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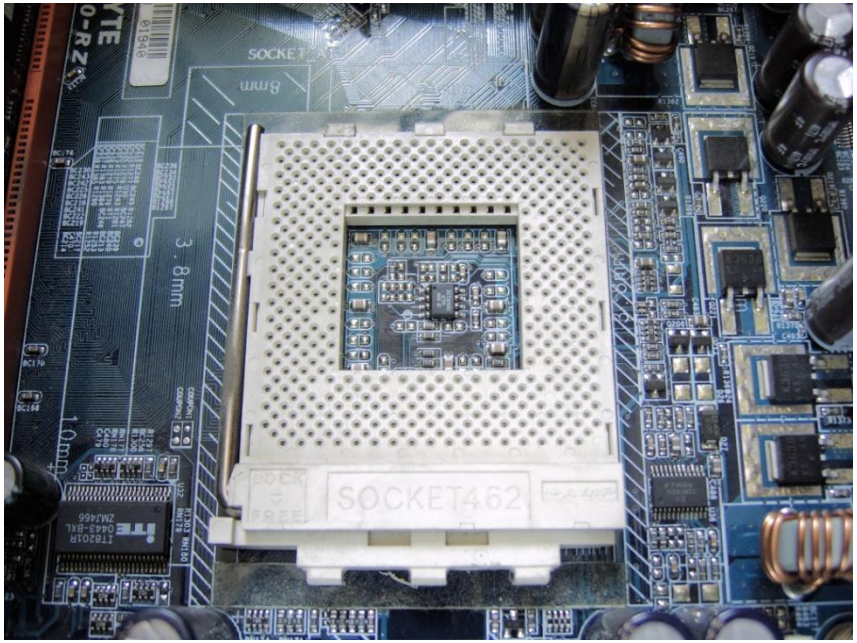
GIGABYTE

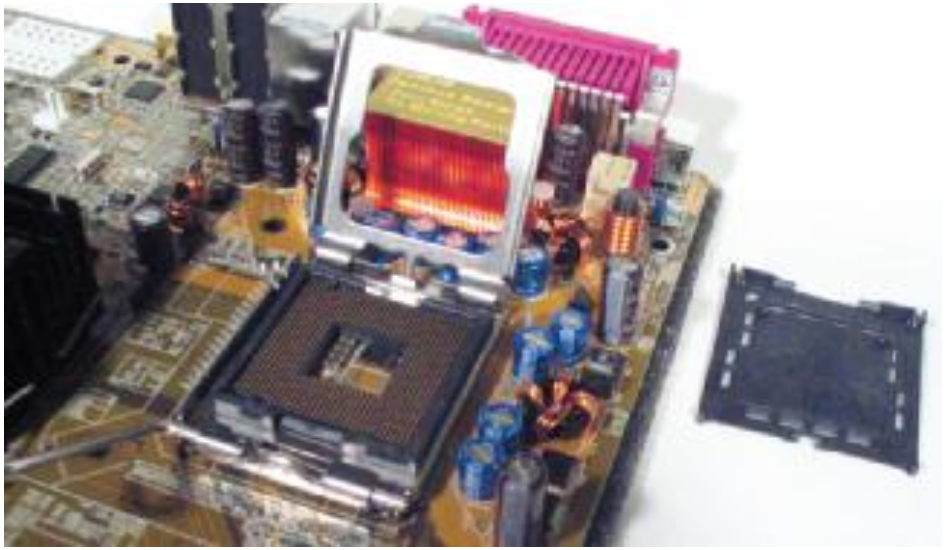
ASUS

Main Socket Types

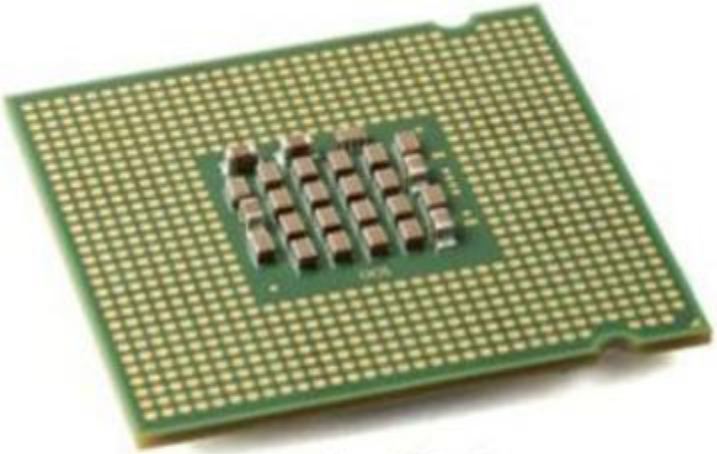
PGA (Pin Grid Array) socket



LGA (Land Grid Array) Socket



Slot : Socket



LGA

PGA

Intel Core i3

Main Socket Types

- A **pin grid array (PGA) socket**
  - Early Intel processors used PGA sockets, but they caused problems because the small delicate pins on the processor were easily bent as the processor was installed in the socket.
- A **land grid array (LGA) socket**
  - LGA sockets generally give better contacts than PGA sockets, and the processor doesn't have the delicate pins.

Processor Frequency

- The speed at which the processor operates internally is called the **processor frequency or clock rate.**
- For example:
  - 3.2 GHz

Categories of processors currently used

- **32-bit processors:**
  - *These older processors are known as **x86 processors because Intel*** used the number 86 in the model number of these processors. 32 bit OS
- **64-bit processors:**
  - *Intel makes several 64-bit processors for workstations or servers that* use fully implemented 64-bit processing.

QUIZ

Which CPU feature enables the microprocessor to support running multiple operating systems at the same time?

A. Clock multiplying

B. Caching

C. Pipelining

D. Virtualization support

Phoenix TrustedCore(tm) Setup Utility

Advanced

Advanced Processor Configuration		Item Specific Help
CPU Mismatch Detection:	[Enabled]	When enabled, a VMM (Virtual Machine Monitor) can utilize the additional hardware capabilities provided by Vanderpool Technology.
Core Multi-Processing:	[Enabled]	
Processor Power Management:	[Disabled]	
Intel(R) Virtualization Technology	[Enabled]	
Execute Disable Bit:	[Enabled]	
Adjacent Cache Line Prefetch:	[Disabled]	
Hardware Prefetch:	[Disabled]	
Direct Cache Access	[Disabled]	If this option is changed, a Power Off-On sequence will be applied on the next boot.
Set Max Ext CPUID = 3	[Disabled]	

F1 Info ↑↓ Select Item -/+ Change Values F9 Setup Defaults
Esc Exit ← Select Menu Enter Select ▶ Sub-Menu F10 Save and Exit

QUIZ

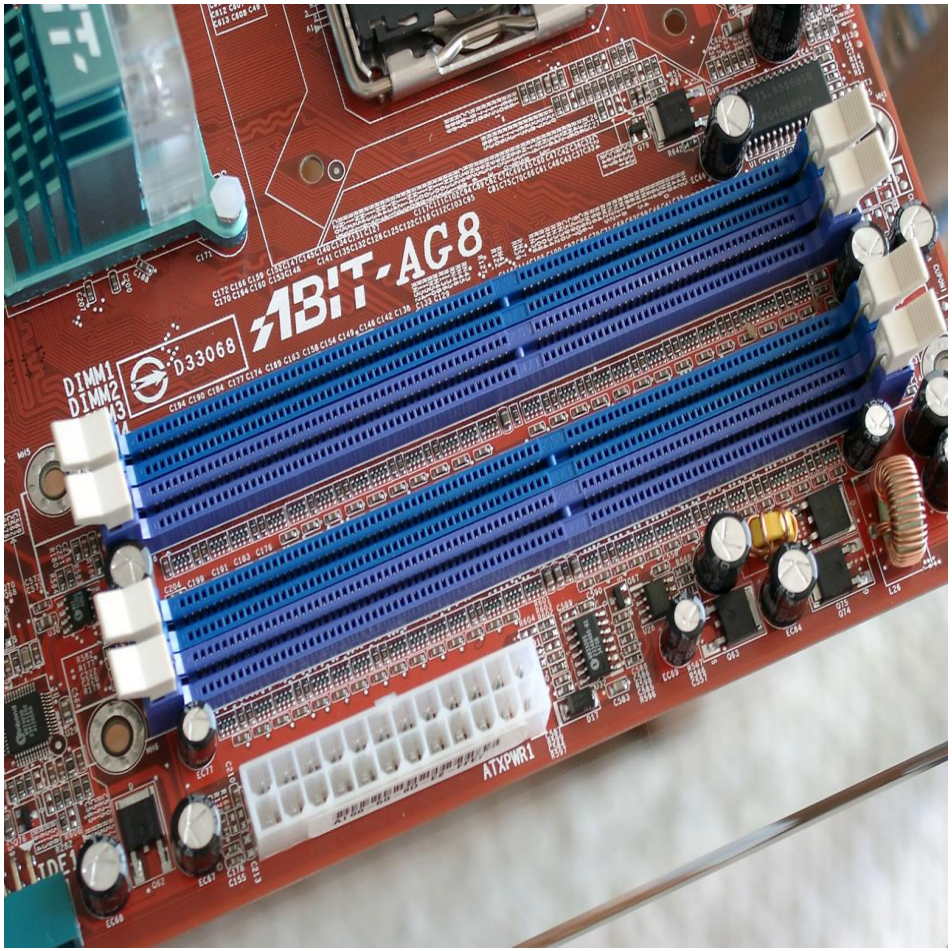
True or False:

1. Speakers can be considered as an input and output device.
2. Parallel ports are more reliable than Serial ports.
3. LGA socket CPUs are easy to fix compared with PGA socket CPUs.
4. The southbridge chipset directly communicates with CPU.
5. Intel Celeron processors has more cache memory for better performances.

MEMORY SLOT

- **Memory slots**
  - DIMM slots for DRAM. (DDR RAM)

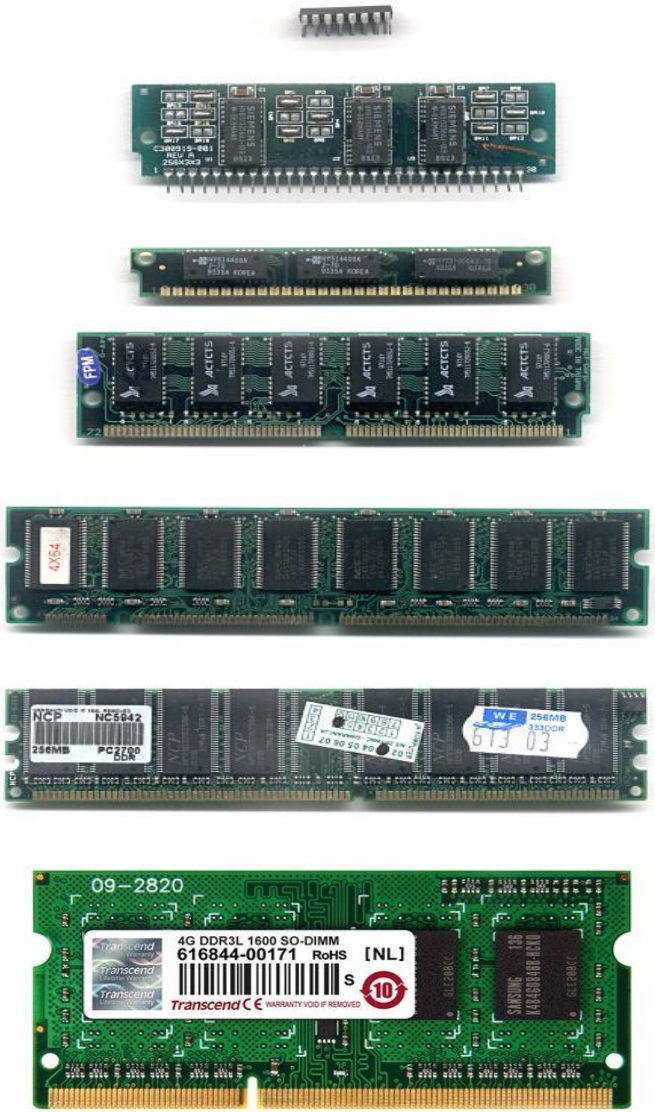
RAM



Random Access Memory (RAM)

- Random Access Memory (RAM) temporarily holds data and instructions as the CPU processes them
- Memory modules used on a motherboard currently are made of dynamic RAM /DRAM
- DRAM loses its data rapidly, and the memory controller must refresh it several thousand times a second.

Different types of DRAM



1. DIP - Dual in-line Package
2. SIPP - Single In-line Pin Package
3. SIMM (30-pin) - Single In-line Memory Module
4. SIMM (72-pin) – Single In-line Memory Module
5. DIMM (168-pin) - Dual In-line Memory Module
6. DDR DIMM (184-pin) - Double Data Rate Dual In-line Memory Module
7. SO-DIMM – Simple Outline Dual Inline Memory Module

There are variety of RAMs available in the market:
They differ based on capacity and speed.

SDRAM - Synchronous Dynamic RAM

DDR SDRAM, **DDR2, DDR3, DDR4** - Double Data Rate Synchronous Dynamic RAM

DDR2

DDR3

HDD **SSD**

Parameter	SSD	HDD
Abbreviation for	Solid State Drive	Hard Disk Drive
Cost	Costlier than HDD	Cheaper
I/O Performance	6000 io/s	400 io/s
Weight	Lighter than HDD	Heavier than SDD
Durability	More durable	Less durable
Energy consumption	Less consumption	Higher consumption
Noise	Very Low	Produces some noise due to moving parts
Technology	Integrated circuit using Flash memory	**Mechanical parts including Spinning disks, or platters**
Target devices	Modern Laptops	Computers and PCs

Partition

QUIZ

Name the following devices:

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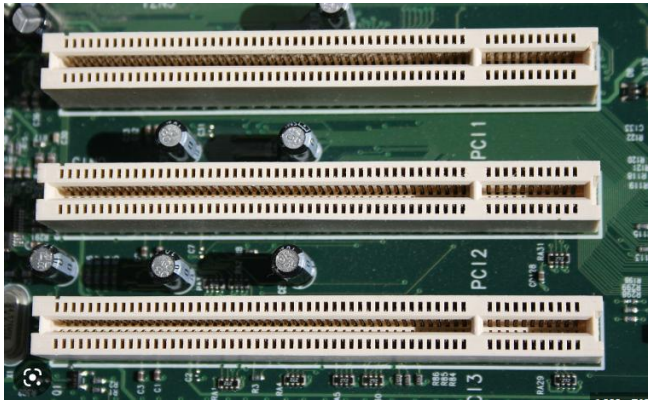

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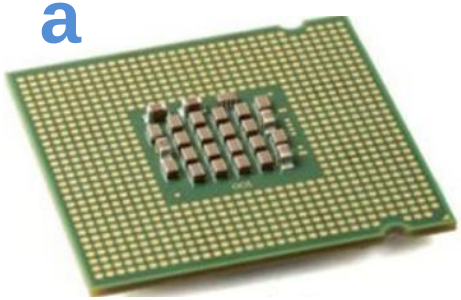

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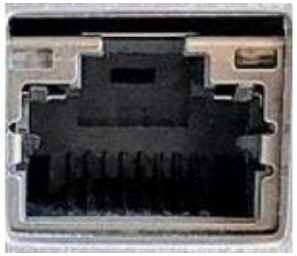

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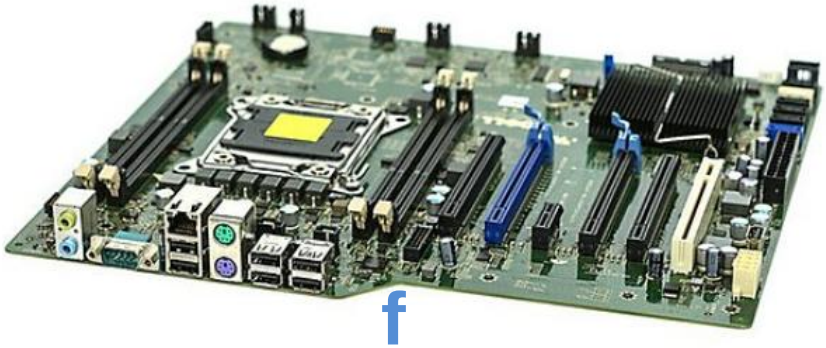

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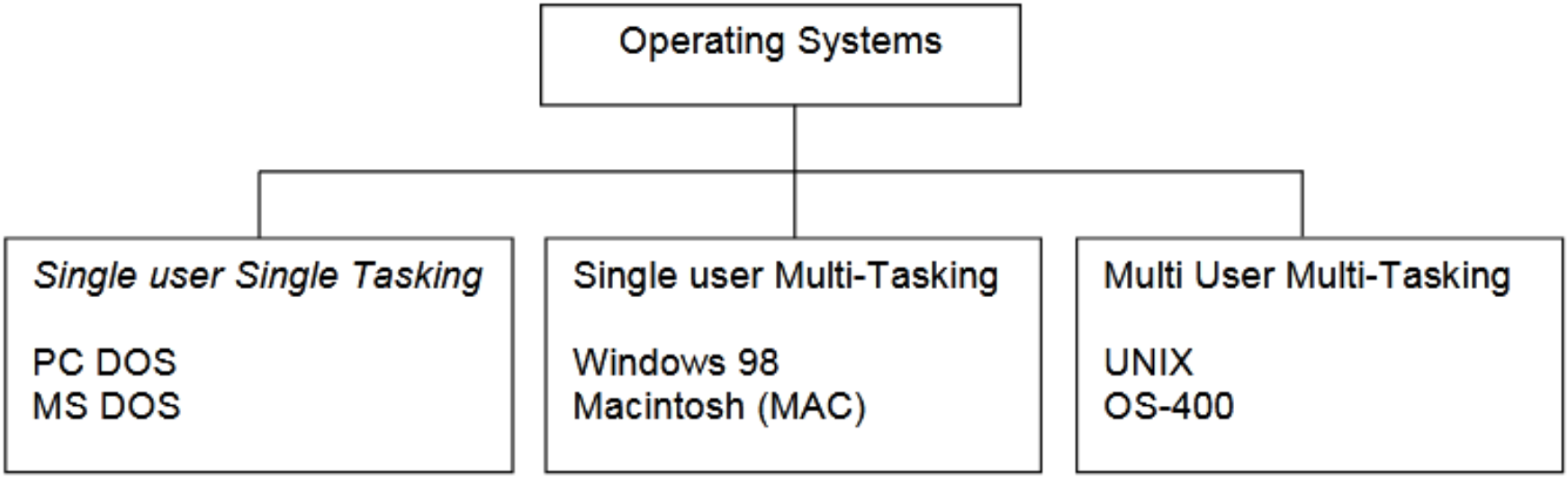
(ROM, PSU, Motherboard, RAM, CPU, COM1, RJ45, PCI slots, ISA slots, SSD)

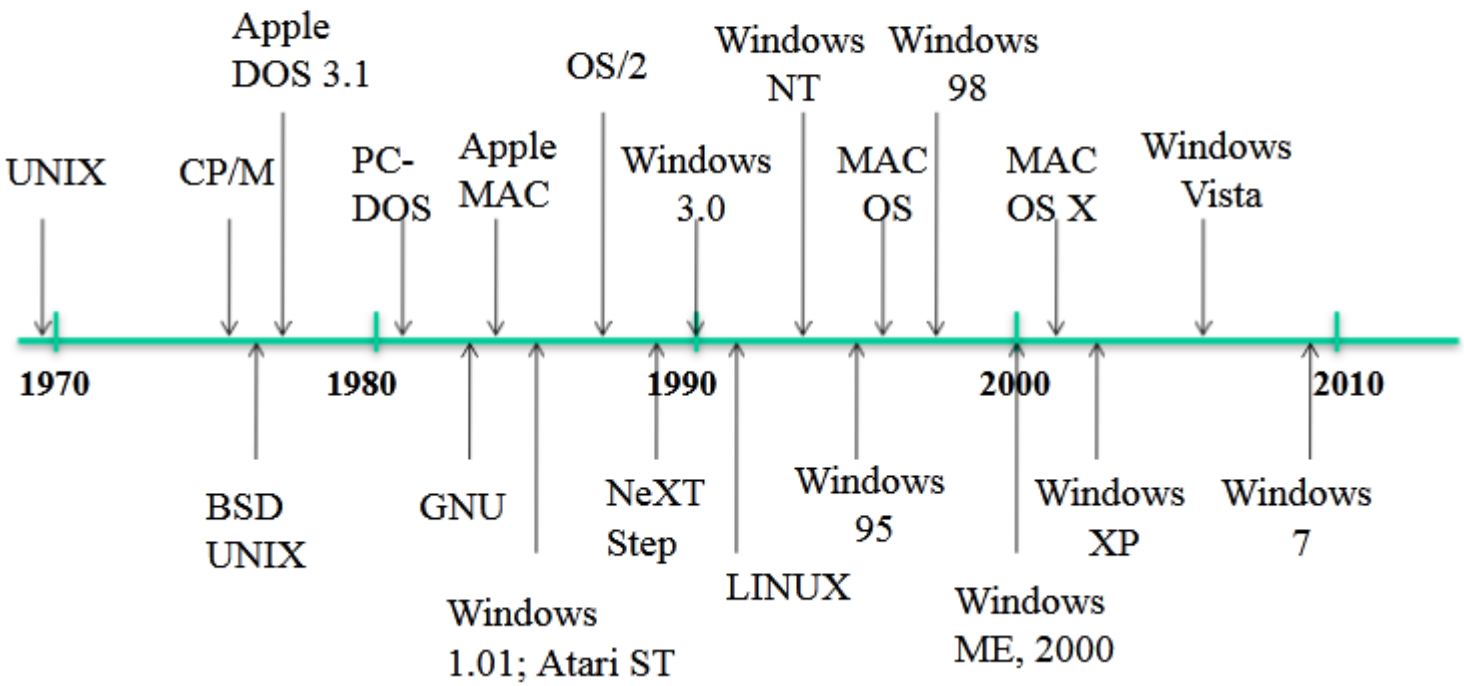
Types of Operating Systems



Note:- In addition to the above now Network Operating Systems (NOS) are also available, which runs on a group of computers that networked for communication and resource sharing. These may be client-server systems or distributed systems. E.g., MS-Window 2000 (Advanced Server), MS-Windows Server 2003

Desktop O/S – Win 98, 2000 prof, XP, 7, 8, 10
Server O/S – Win 2000 ser, 2013, 2008, 2008 r2, 2012, 2012 r2, 2016

Timeline of OS Milestones



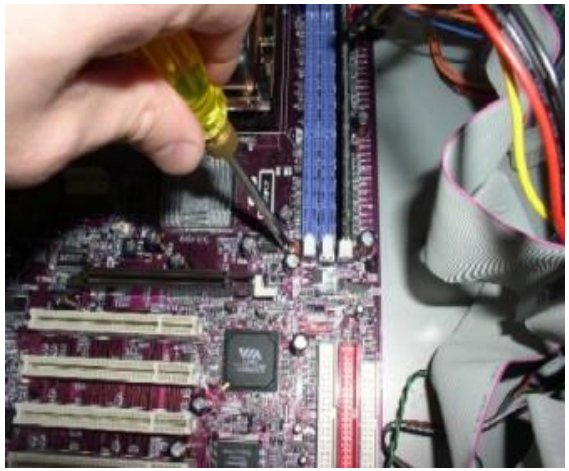
PC Assembly

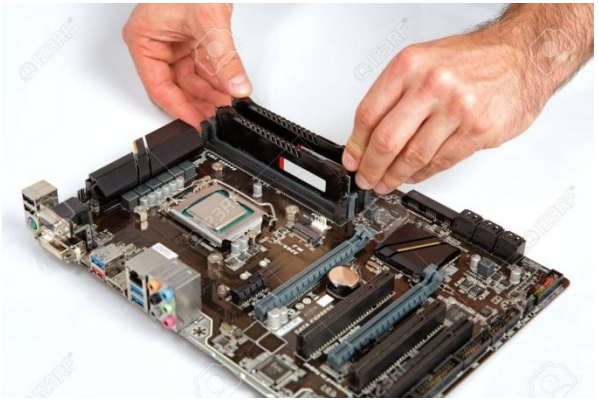
Things to Prepare before Assembly

- Anti-static wrist strap
- Set of screwdrivers and pliers
- Piece of cloth
- CPU Thermal compound (recommended)

Step 1: Installing the motherboard

- Make sure you have all the components in place and a nice, clean and big enough place to work with.
- Put your antic-static wrist strap on to prevent your components from getting affected. Make sure your hands are clean before starting.
- Unscrew and open the side doors of the cabinet
- Lay the cabinet on its side
- Put the motherboard in place
- Drive in all the required screws on the motherboard screw mounting points.





Step 2: Installing the CPU

- CPU is the heart of a computer so make sure you handle it properly and do not drop it or mishandle it.
- <u>DO NOT</u> touch the pins.
- Get hold of your motherboard and CPU manual
- Lift the CPU lever on the motherboard
- Place the CPU properly on the motherboard
- Pull down the lever to secure the CPU in place

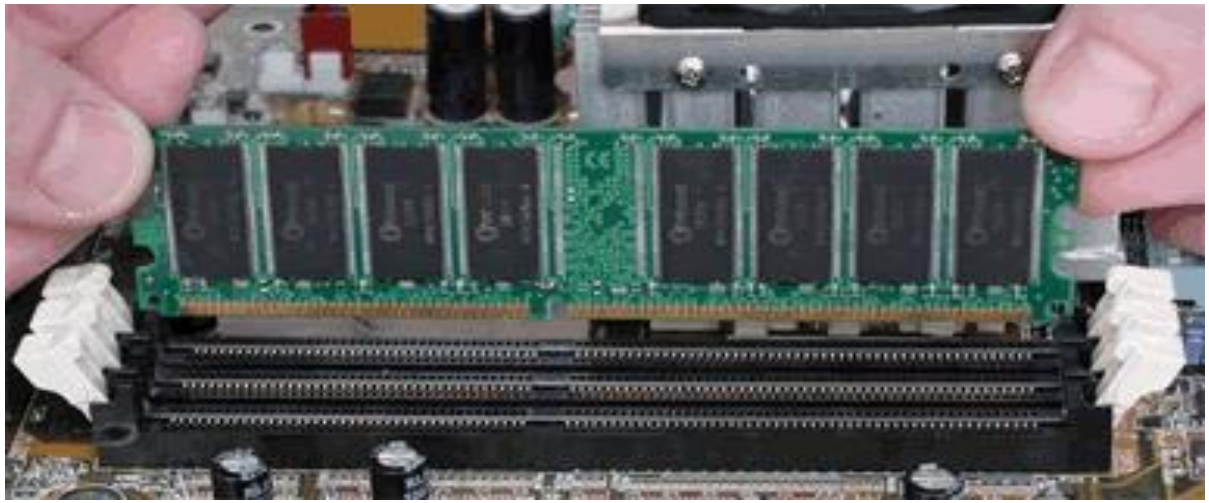
AMD Processor

Step 3: Installing the heat sink

- After installing the processor we proceed to installing the heat sink. There are different kinds of heat sinks that are bundled with the processor and each has a different way of installation.
- Place the heat sink on the processor
- Secure the heat sink with the lever or screws.
- After this you will need to connect the cable of the heat sink fan on the motherboard.

Step 4: Installing the RAM

- The newer RAMs ie. DDR RAMs are easy to install as you don’t have to worry about placing which side where into the slot.
- DDR RAM terminals are split into 2 lengths to fit in to the RAM connector.
- Push down the RAM into the slot
- Make sure the both the clips hold the RAM properly

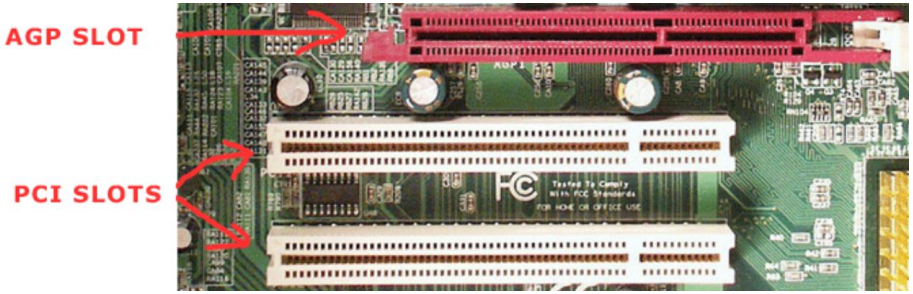


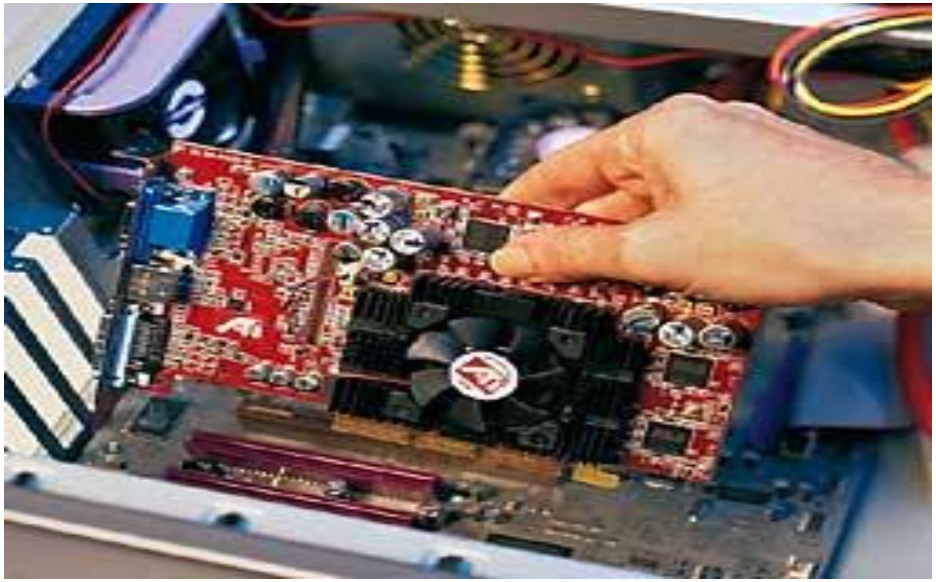
Step 5: Installing the power supply

- We will now install the power supply as the components we install after this will require power cables to be connected to them.
- Place the PSU into the case bracket
- Put the screws in place tightly

Step 6: Installing the video card

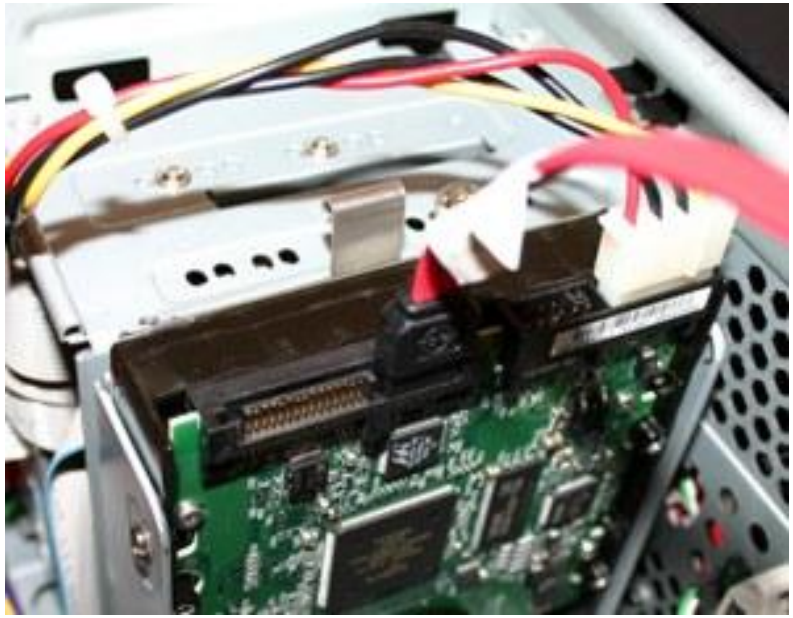
- First you will need to find out whether your video card is AGP or PCI-E.
- Remove the back plate on the case corresponding to the graphics card
- Push the card into the slot
- Secure the card with a screw





Step 7: Installing the hard disk

- Hard disks are breakable and needs to handled carefully.
- Place the hard drive into the bay
- Secure the drive with screws
- Connect the power cable from PSU to hard disk
- Connect the PATA (IDE) or SATA cable from motherboard into the drive



Step 8: Installing optical drive

- The installation an optical drive is exactly similar to a hard drive.
- Place the optical drive into the bay
- Drive in the screws
- Connect the power cable and data cable

Step 9: Connecting various cables

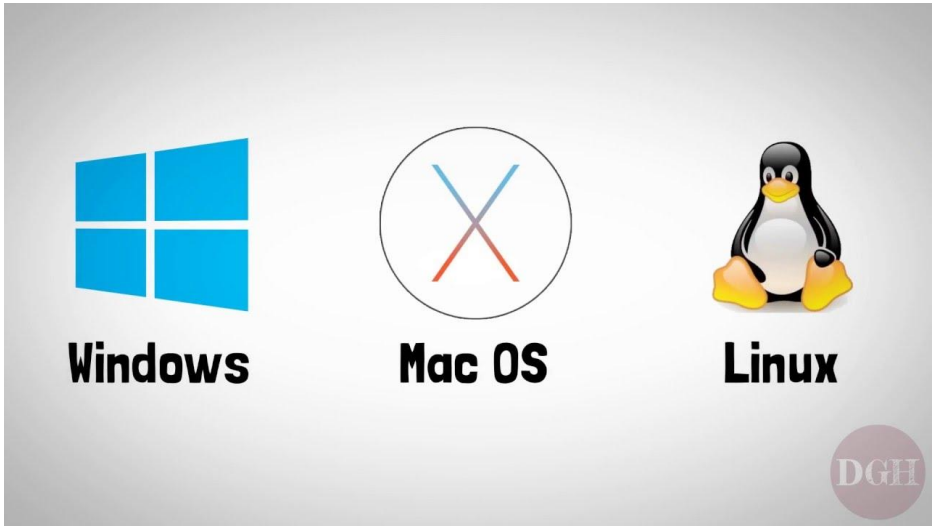
- You will need to consult your motherboard manual for finding the appropriate port for connecting various cables at the right places on the motherboard.
- Connect the large ATX power connector to the power supply port on your motherboard
- Next get hold of the smaller square power connector which supplies power to the processor and connect it to the appropriate port by taking help from your motherboard manual
- Connect the ATX case cables for power, reset button in the appropriate port of the motherboard
- Connect the front USB/audio panel cable in the motherboard
- Plug the cable of ATX case cooling fans

Step 10:

- Get the rest of the PC components like monitor, keyboard, mouse, speakers etc. which we will connect now.
- Connect the VGA cable of the monitor into the VGA port
- If mouse/keyboard are PS/2 then connect them to PS/2 ports or else use the USB port
- Connect the speaker cable in the audio port
- Plug in the power cable from PSU into the UPS
- Also plug in the power cable of the monitor

Step 11: Installing the OS and drivers

- Insert OS CD/DVD into CD/DVD rom drive.
- Set the first boot device to CD/DVD drive in BIOS
- Pop in the OS disk
- Reboot the PC
- Install the OS
- Install drivers from motherboard CD (applicable only to Windows OS)



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